A MORPHISM INDUCES THE EXPECTED MAP ON RESIDUAL GERBES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3100-3113 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 3115-3123, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 38 of queue_stacks-properties.json (pb=415, stmt 3100-3113, proof 3115-3123).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-monomorphism-into-point: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2759–2764. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-residual-gerbe-points: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3054–3060. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $x \in |\mathcal{X}|$ with image $y \in |\mathcal{Y}|$. If the residual gerbes $\mathcal{Z}_x \subset \mathcal{X}$ and $\mathcal{Z}_y \subset \mathcal{Y}$ of x and y exist, then f induces a commutative diagram*

$$\begin{array}{ccc} \mathcal{X} & \longleftarrow & \mathcal{Z}_x \\ {\scriptstyle f}\downarrow & & \downarrow \\ \mathcal{Y} & \longleftarrow & \mathcal{Z}_y \end{array}$$

Proof. Choose a field k and a surjective, flat, locally finitely presented morphism $\mathrm{Spec}(k) \to \mathcal{Z}_x$. The morphism $\mathrm{Spec}(k) \to \mathcal{Y}$ factors through $\mathcal{Z}_y$ by Lemma lemma-residual-gerbe-points. Thus $\mathcal{Z}_x \times_{\mathcal{Y}} \mathcal{Z}_y$ is a nonempty substack of $\mathcal{Z}_x$ hence equal to $\mathcal{Z}_x$ by Lemma lemma-monomorphism-into-point. □

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.